

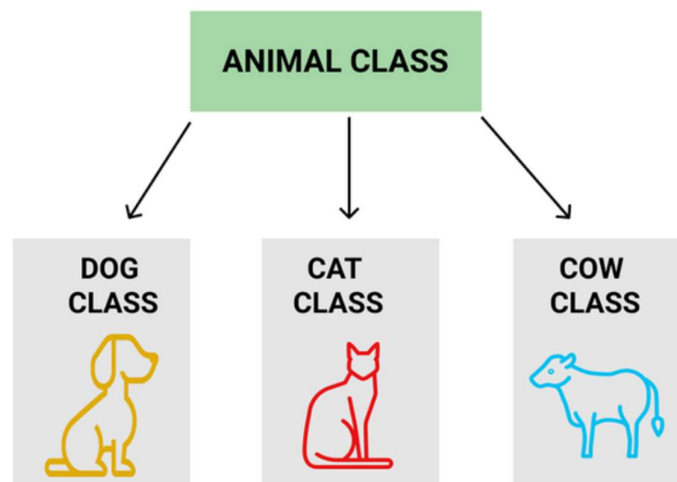
# Inheritance in Java

## ✚ What is Inheritance :-

- Java Inheritance is a fundamental **concept in OOP(Object-Oriented Programming)**.
- where **one class (child/subclass)** can use the properties and methods of **another class (parent/superclass)**.
- It helps in **code reusability** and makes programs easier to maintain.
- **Inheritance** allows a class to **borrow features** from another class.
- Example: A **Dog** class can inherit from an **Animal** class to reuse its properties like eat() or sleep().

## ✚ Key Points in inheritance :-

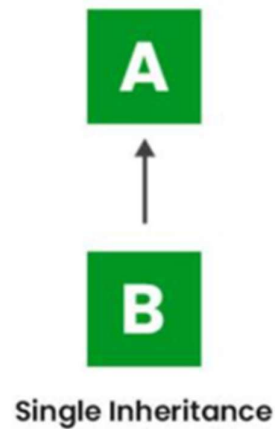
- The **extends** keyword is used to inherit a class.
- The **child class** can access **variables, methods, and constructors** of the parent class.
- Supports **single, multilevel, and hierarchical** inheritance in Java.
- Java does not support **multiple inheritance** using classes (to avoid confusion).



## + Types of Inheritance :-

### + Single Inheritance :-

- In single inheritance, a sub-class is derived from **only one super class**.
- It inherits the **properties** and **behavior** of a single-parent class.



### ➤ Syntax :-

```
public class A {  
    ...  
}
```

```
public class B extends A {  
    ...  
}
```

## Implementation of single inheritance :-

Input :-

```
package Oops;

public class Child extends Parent{

    void second()
    {
        System.out.println("Child Property");
    }
    public static void main(String[] args) {

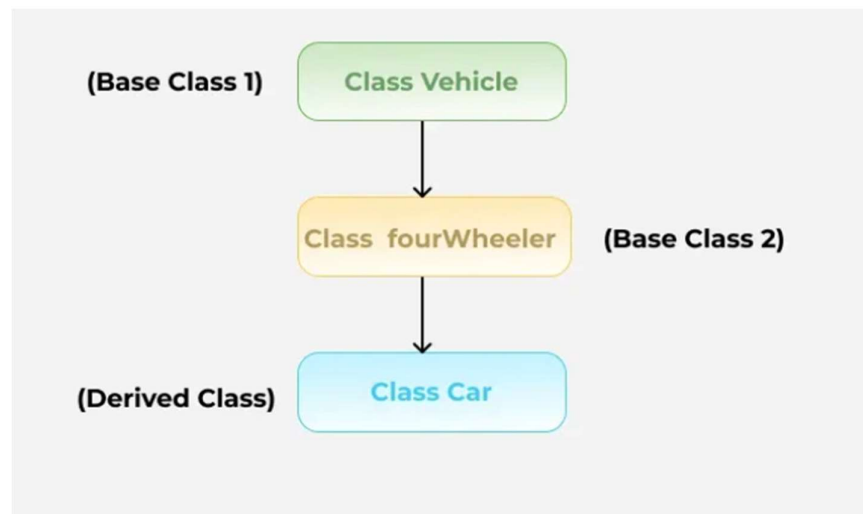
        Child ob=new Child();
        ob.second();
        ob.first();
    }
}
1 package Oops;
2
3 public class Parent{
4
5     void first()
6     {
7         System.out.println("Parent Property");
8     }
9 }
10
```

Output :-

```
Child Property
Parent Property
```

## ✚ Multilevel Inheritance :-

- In Multilevel Inheritance, a derived class will be inheriting a **base class** and as well as the **derived class** also acts as the base class for other classes.
- This creates a **chain or hierarchy of inheritance**, with **multiple levels of parent-child relationships**.



## ➤ Syntax :-

```
class A {  
  //  
}
```

```
class B extends A {  
  //  
}
```

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```
class C extends B {  
  //  
}
```

## Implementation of Multilevel inheritance :-

### Input :-

```
1 package Oops;
2
3 public class multilevel_child extends multilevel_parent {
4
5     void cp()
6     {
7         System.out.println("Child class");
8     }
9     public static void main(String[] args) {
10         multilevel_child obj=new multilevel_child();
11         obj.cp();
12         obj.mp();
13         obj.mgp();
14     }
15 }
```

```
1 package Oops;
2
3 public class multilevel_parent extends multilevel_grandparent{
4
5     void mp()
6     {
7         System.out.println("Parent Class");
8     }
9 }
```

```
1 package Oops;
2
3 public class multilevel_grandparent {
4
5     void mgp()
6     {
7         System.out.println("Grandparent class");
8     }
9 }
```

### Output :-

---

```
Child class
Parent Class
Grandparent class
```

## Example

### Input :-

```
1 package Oops;
2 class institute{
3     void details()
4     {
5         System.out.println("-----Institute-----");
6
7         System.out.println("Fortune Cloud");
8         System.out.println("Cravita technology");
9         System.out.println("Mind Space");
10    }
11 }
12 public class courses extends institute{
13
14     public static void main(String[] args) {
15         courses ob=new courses();
16         ob.details();
17         ob.info();
18     }
19
20     void info()
21     {
22         System.out.println("-----Courses-----");
23         System.out.println("Java");
24         System.out.println("Python");
25         System.out.println("Dot Net");
26         System.out.println("Mern Stack");
27     }
28 }
```

### Output :-

```
-----Institute-----
Fortune Cloud
Cravita technology
Mind Space
-----Courses-----
Java
Python
Dot Net
Mern Stack
```